

CMPT 354 Mini Project

Step 1: Project Specifications

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1 Project Overview

The goal of this project is to build a database application for a community library. The database will keep one consistent record of the library's collection, loans, fines, events, rooms, members, personnel, donations, volunteers, help requests, and items that may be acquired in the future. The system should help library users find and use services while allowing personnel to maintain accurate records and enforce library rules.

For this project, the library is treated as one location with several social rooms. A person may visit the library without an account, but an active member account is required to borrow an item, register for an event, donate an item, volunteer, or submit a help request through the application.

2 Users of the System

The system will support the following users:

- **Visitors** may search the collection and view public event information.
- **Members** may perform all visitor actions. They may also borrow and return items, view their loans and fines, register for events, offer donations, apply to volunteer, and ask a librarian for help.
- **Library personnel** may add and update collection records, process loans and returns, record damaged or lost items, manage events and registrations, review donations and proposed items, approve volunteers, and respond to help requests.
- **Administrators** may perform personnel actions and maintain employee records, library policies, and user access permissions.

Each user will be allowed to view or change only the information required for their role. For example, ordinary members must not be able to view private personnel records or change fines.

3 Information Stored by the Database

3.1 Members

For each member, the library will store a unique membership number, full name, email address, phone number, mailing address, date joined, membership status, and membership expiry date. A membership may be active, suspended, expired, or closed. The database will also preserve the member's loan, fine, event, donation, volunteer, and help-request history.

Each member email address is required and must be unique among member records.

3.2 Personnel

For each employee, the library will store a unique employee number, name, contact information, job title, hire date, employment status, and supervisor when applicable. Job titles may include librarian, event coordinator, cataloguer, and administrator. Only authorized administrators may view or update private personnel information.

Each employee email address is required and must be unique among employee records.

3.3 Library Collection

The collection may contain print books, online books, magazines, scientific journals, music records, films, and other materials. The database will separate the description of a title from the individual copies owned or licensed by the library. This is necessary because several copies of the same title may exist, and each copy may have a different location, condition, or availability.

For every title, the library will store a unique catalogue number, title, material type, creator or author, publisher, publication year, language, subject, description, and a standard identifier such as an ISBN or ISSN when one exists. A standard identifier must be unique when it is present. Each creator will have a unique creator number and a name. A title may have several creators, and one creator may be connected to several titles. Individual issues of magazines and journals will also record their volume, issue number, and publication date; the combination of catalogue number, volume, and issue number uniquely identifies an issue.

Every lendable unit will have a unique internal unit number and a current status. Its status may be available, on loan, reserved for processing, damaged, lost, or withdrawn. A physical copy will also have a unique barcode, shelf location, acquisition date, and condition. A digital licence will instead have a unique licence number, access link, and licence limit. The number of simultaneous digital loans must never exceed the licence limit.

3.4 Loans, Returns, and Fines

For each loan, the database will store a unique loan number, the member, the borrowed lendable unit, the employee who processed the transaction when applicable, the date and time borrowed, the due date, and the actual return date. The return date remains empty while the loan is active.

The normal loan periods are 21 days for print books, 14 days for online books, and 7 days for magazines, journal issues, records, and films. An active member may have at most five active loans. A physical copy cannot be included in more than one active loan at the same time. A member with an overdue item or at least \$20 in unpaid fines cannot begin another loan. Digital access ends automatically on its due date.

If a physical item is returned late, the member is charged \$0.25 for each day after the due date. For each fine, the database will store a unique fine number, the amount, date assessed, reason, payment status, payment date, and the loan that caused the fine. Fine amounts cannot be negative. Lost or seriously damaged items may receive an additional replacement charge recorded with the same loan.

3.5 Rooms and Events

For each social room, the library will store a unique room number, room name, floor, capacity, accessibility information, and availability status. Events may include book clubs, author talks, workshops, art shows, and film screenings.

For each event, the database will store a unique event number, title, event type, description, start time, end time, room, maximum attendance, organizing employee, and event status. An event may be scheduled, cancelled, or completed. All events are free. The maximum attendance cannot be greater than the capacity of the assigned room, and two events cannot use the same room during overlapping times.

Every event will be recommended for one or more audiences, such as children, teens, adults, seniors, families, beginners, or researchers. An audience may be recommended for many events. The recommendation is informational and does not prevent other members from attending unless an event has a stated age or safety restriction.

For each event registration, the system will record a unique registration number, the member, event, registration time, registration status, and whether the member attended. A member may register for the same event only once. Registration is allowed only while the event is scheduled and has remaining capacity.

3.6 Donations and Possible Future Items

The library will keep records of materials that members offer to donate and materials that personnel may purchase in the future. Each proposed item will include a unique proposal number, title, creator, material type, publication information, source of the proposal, date submitted, condition when applicable, notes, and current decision status. The source may be a member donation, a member request, or a personnel purchase suggestion.

A proposal may be pending, under review, accepted, or rejected. When a member submits a donation, the member will be recorded as the donor. A reviewing employee and decision date will be recorded after a decision is made. Accepting a proposal does not immediately make it borrowable; personnel must first create the catalogue and copy records and mark the copy as available. Rejected proposals remain in the history so that the same item is not reviewed repeatedly without a reason.

3.7 Volunteering and Help Requests

A member may submit a volunteer application for a particular event or for general library work. The application will record a unique application number, the member, application date, area of interest, availability, relevant skills, reviewing employee, decision, assignment notes, and optional event. An application may be pending, approved, rejected, or completed. The reviewing employee, assignment notes, and event may remain empty until applicable. Approval must be recorded by an employee before the member is given a volunteer assignment.

A member may also ask a librarian for help. Each help request will record a unique request number, member, request category, question or description, submission time, priority, status, assigned librarian, response, and completion time. The assigned librarian, response, and completion time may remain empty until applicable. A request may be open, assigned, resolved, or closed. A request cannot be marked resolved without a response and a completion time.

4 Required Application Operations

The completed application must allow an authorized user to perform the following operations while preserving all library rules:

1. Search for an item by title, creator, subject, material type, catalogue number, ISBN/ISSN, or availability, and view the location or online access information that the user is permitted to see.
2. Borrow an available item after checking the member's status, current number of loans, overdue items, unpaid fines, and the availability of the physical copy or digital licence.
3. Return an active loan, update the copy's condition and availability, and assess a fine when the return is late, lost, or damaged.
4. Submit and review a possible donation, and convert an accepted donation into catalogue and copy records after processing.
5. Search for events by title, type, date, audience, or availability.
6. Register a member for a free event after checking event status, room capacity, and duplicate registration.
7. Submit and review a volunteer application and, when approved, record its assignment or related event.
8. Submit, assign, answer, and close a request for help from a librarian.

Personnel must also be able to view operational reports, including currently borrowed and overdue items, unpaid fines, item availability, room schedules, event registrations and attendance, pending donations, volunteer applications, and unresolved help requests.

5 Data Integrity and Record-Keeping Rules

The following rules apply throughout the database:

- Membership numbers, employee numbers, creator numbers, catalogue numbers, lendable-unit numbers, physical-copy barcodes, digital-licence numbers, room numbers, loan numbers, fine numbers, event numbers, registration numbers, proposal numbers, volunteer-application numbers, and help-request numbers must be unique. Member and employee email addresses are also unique within their respective record types. A periodical issue is unique by its catalogue number, volume, and issue number, and a member-event pair is unique in event registrations.

- Required names, titles, dates, and status values cannot be empty, and all stored dates must follow a consistent format.
- A due date must be after its borrowing date, an event's end time must be after its start time, and a payment or completion date cannot be earlier than the record that caused it.
- A loan must refer to an existing member and lendable unit. A fine must refer to an existing loan. An event must refer to an existing room and organizing employee. Registrations, item proposals, volunteer applications, and help requests must refer to their related existing records.
- Status values must come from the allowed values described above. Counts, capacities, licence limits, and monetary amounts cannot be negative.
- Important history such as completed loans, paid fines, past events, reviewed proposals, and resolved requests will not be deleted during normal operation. If a record is no longer active, its status will be changed so that reports remain accurate.
- The responsible employee will be recorded for employee-processed loans, event organization, proposal review, volunteer review, and help-request assignment whenever the corresponding workflow applies.

6 Scope Limitations

The first version of the system will not handle employee payroll, paid event tickets, inter-library loans, room rentals, or the purchase of commercial digital licences. It will only keep the information needed to support the library services and application operations described above.

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Step 2: Entity–Relationship Model

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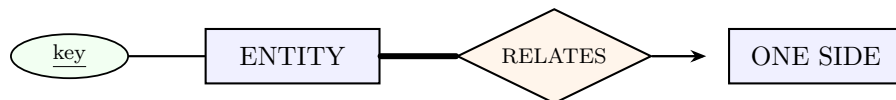
1 Model and Notation

The following E/R model implements the library requirements specified in Step 1. The model separates catalogue descriptions from the individual physical copies and digital licences that can be borrowed. It also represents loans and fines, personnel, events and registrations, future-item proposals, volunteer applications, and help requests.

The diagrams use the notation introduced in class:

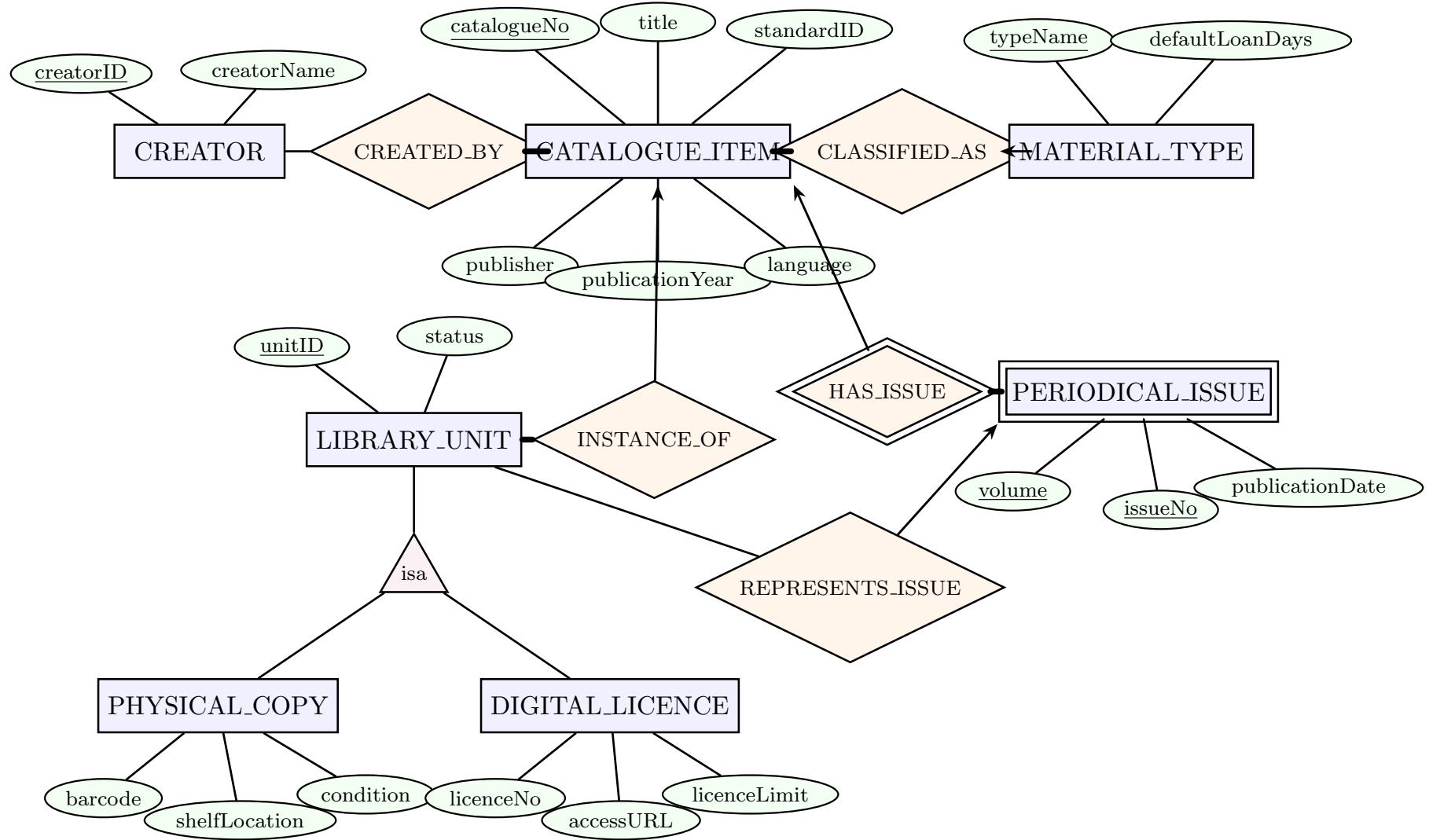
- a rectangle is an entity set, an oval is an attribute, and a diamond is a relationship set;
- an underlined attribute is a primary key;
- an arrow points toward the “one” side of a relationship;
- a thick line represents total participation;
- a triangle represents an *isa* relationship; and
- a double rectangle and double diamond represent a weak entity set and its identifying relationship.

Only the most important attributes are drawn as ovals to keep the diagrams readable. Section 7 gives the complete attribute set and is part of the E/R definition. Attributes marked **AK** there are additional candidate keys, but only the selected primary key is underlined in the diagrams, as required by the class notation.



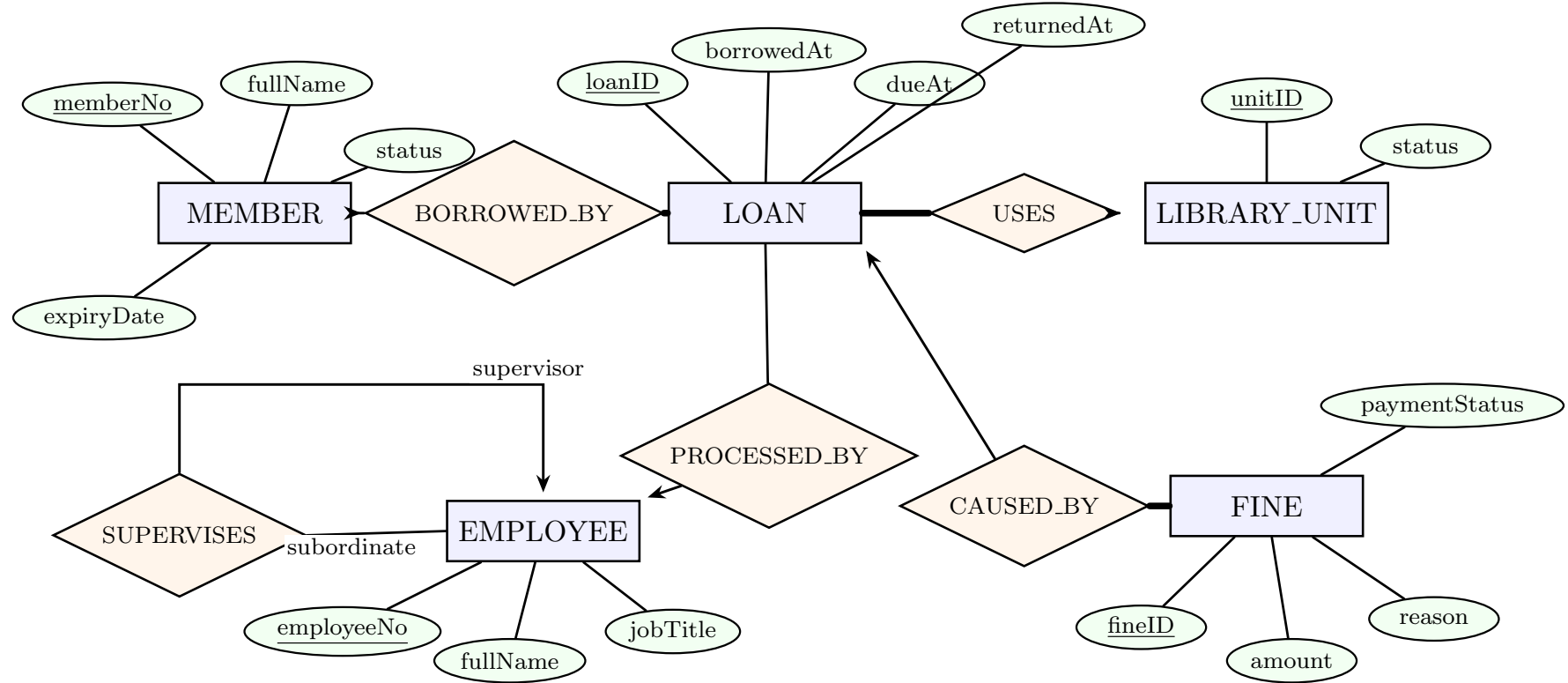
In the example above, every ENTITY participates in exactly one RELATES relationship with ONE SIDE. The thick line gives the minimum of one, and the arrow gives the maximum of one.

2 Catalogue and Lendable Units



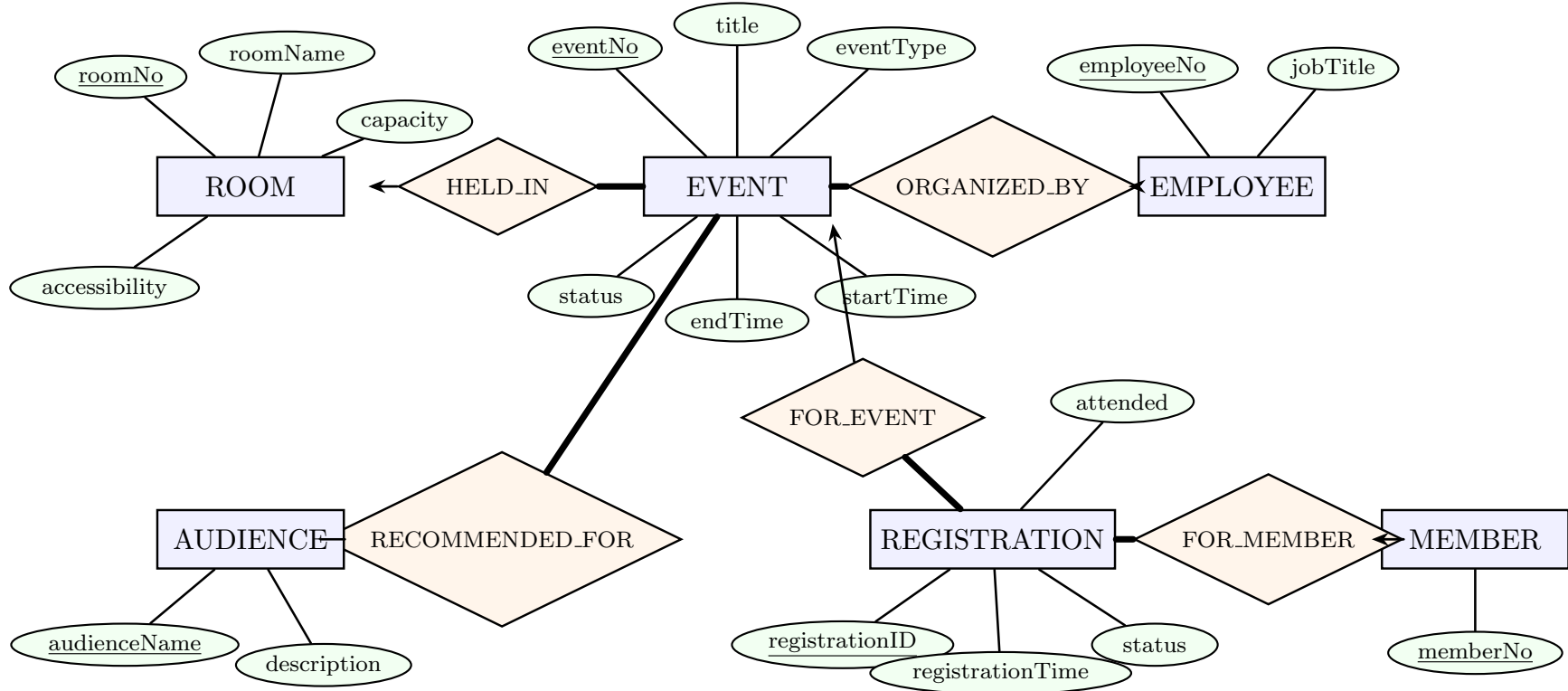
Cardinality and participation. CREATED_BY is many-many, and every catalogue item has at least one creator. Every catalogue item has exactly one material type. Every library unit is an instance of exactly one catalogue item. LIBRARY_UNIT is the superclass of PHYSICAL_COPY and DIGITAL_LICENCE; the subclasses are disjoint and total. PERIODICAL_ISSUE is weak: its key is (*catalogueNo*, *volume*, *issueNo*). REPRESENTS_ISSUE is optional for a unit and is used only for magazine and scientific-journal issues.

3 Members, Personnel, Loans, and Fines



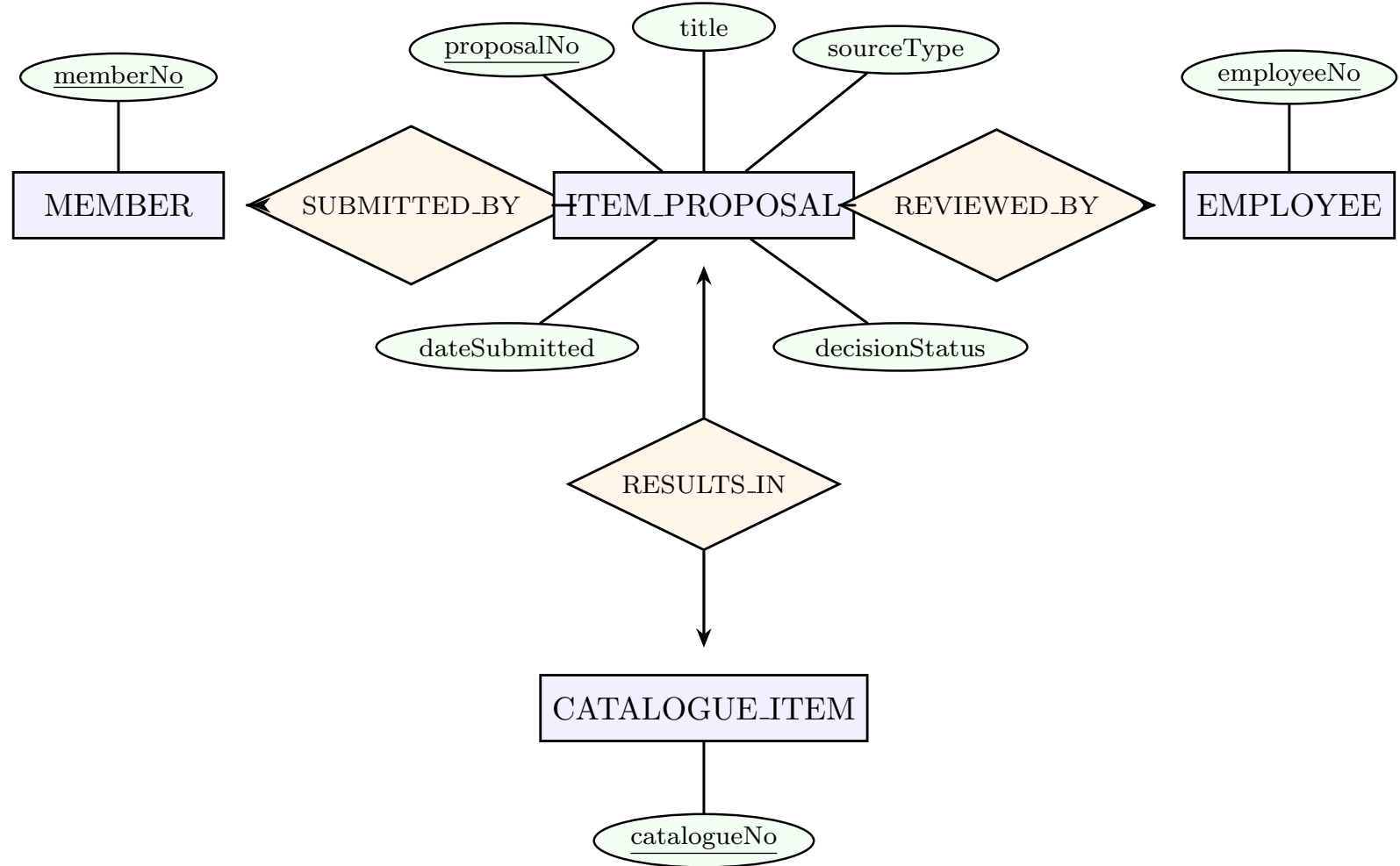
Cardinality and participation. Every loan is for exactly one member and exactly one library unit. A member and a unit may each occur in many loans over time. PROCESSED_BY is optional because a digital loan may be completed without an employee. Every fine is caused by exactly one loan, while one loan may cause zero or more fines. In the self-relationship SUPERVISES, an employee has at most one supervisor but may supervise many employees.

4 Rooms, Events, Audiences, and Registrations



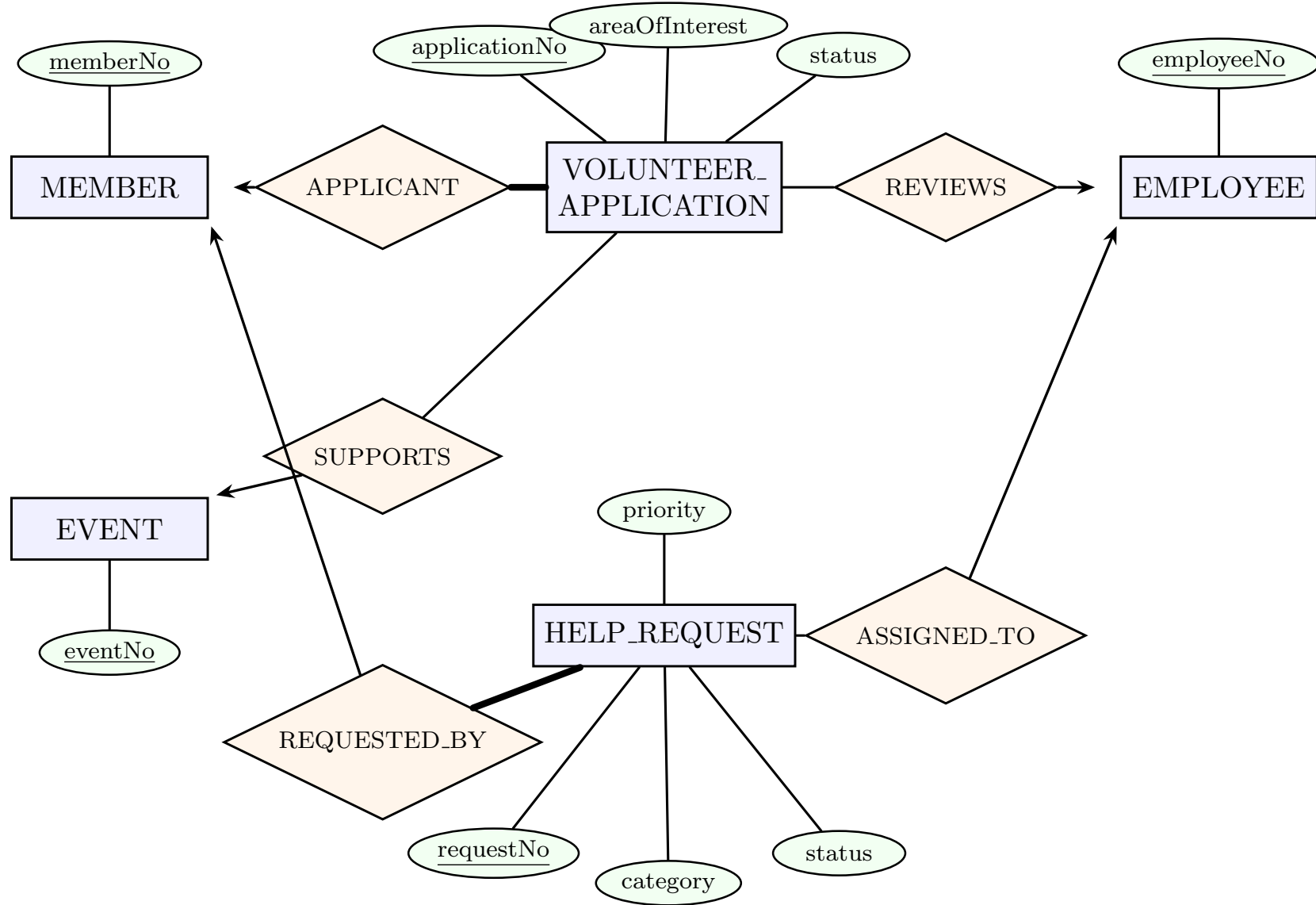
Cardinality and participation. Every event is held in exactly one room and organized by exactly one employee. RECOMMENDED_FOR is many-many, and every event has at least one recommended audience. REGISTRATION is a connecting entity: every registration belongs to exactly one event and exactly one member. The pair (*eventNo*, *memberNo*) is additionally unique, preventing a member from registering for the same event twice.

5 Future-Item and Donation Proposals



Cardinality and participation. SUBMITTED_BY is optional because personnel may enter a purchase suggestion; it is required when *sourceType* is “member donation” or “member request.” A proposal is reviewed by at most one employee and may result in at most one catalogue item. RESULTS_IN is one-one and optional on both sides.

6 Volunteering and Help Requests



Cardinality and participation. Every volunteer application has exactly one member applicant. Its reviewing employee and supported event are optional, and each is limited to at most one. Every help request is submitted by exactly one member and is assigned to at most one employee.

7 Complete Entity and Attribute Dictionary

Entity set	Attributes
MEMBER	<u>memberNo</u> , fullName, email (AK), phone, mailingAddress, dateJoined, status, expiryDate.
EMPLOYEE	<u>employeeNo</u> , fullName, email (AK), phone, jobTitle, hireDate, employmentStatus.
CREATOR	<u>creatorID</u> , creatorName.
MATERIAL_TYPE	<u>typeName</u> , defaultLoanDays. Examples include print book, online book, magazine, scientific journal, record, and film.
CATALOGUE_ITEM	<u>catalogueNo</u> , title, publisher, publicationYear, language, subject, description, standardID (AK when present).
PERIODICAL_ISSUE	Weak entity with partial-key attributes <u>volume</u> and <u>issueNo</u> , plus publicationDate. Its full key also includes the identifying CATALOGUE_ITEM's catalogueNo.
LIBRARY_UNIT	<u>unitID</u> , status. Status is available, on loan, reserved for processing, damaged, lost, or withdrawn.
PHYSICAL_COPY	Subclass of LIBRARY_UNIT; barcode (AK), shelfLocation, acquisitionDate, condition.
DIGITAL_LICENCE	Subclass of LIBRARY_UNIT; licenceNo (AK), accessURL, licenceLimit.
LOAN	<u>loanID</u> , borrowedAt, dueAt, returnedAt.
FINE	<u>fineID</u> , amount, assessedDate, reason, paymentStatus, paymentDate.
ROOM	<u>roomNo</u> , roomName, floor, capacity, accessibility, availabilityStatus.
EVENT	<u>eventNo</u> , title, eventType, description, startTime, endTime, maximumAttendance, status. Events are free, so no price attribute is needed.
AUDIENCE	<u>audienceName</u> , description.
REGISTRATION	<u>registrationID</u> , registrationTime, status, attended.
ITEM_PROPOSAL	<u>proposalNo</u> , title, creator, materialType, publicationInformation, sourceType, dateSubmitted, condition, notes, decisionStatus, decisionDate.
VOLUNTEER_APPLICATION	<u>applicationNo</u> , applicationDate, areaOfInterest, availability, skills, status, assignmentNotes.
HELP_REQUEST	<u>requestNo</u> , category, questionDescription, submittedAt, priority, status, response, completedAt.

8 Relationship Constraints

Relationship	Multiplicity	Participation rule
CREATED_BY	CATALOGUE_ITEM $M : N$ CREATOR	Every item has at least one creator; a creator may be associated with many items.
CLASSIFIED_AS	CATALOGUE_ITEM $N : 1$ MATERIAL_TYPE	Every item has exactly one type.
HAS_ISSUE	PERIODICAL_ISSUE $N : 1$ CATALOGUE_ITEM	Identifying relationship; every issue belongs to exactly one periodical catalogue item.
INSTANCE_OF	LIBRARY_UNIT $N : 1$ CATALOGUE_ITEM	Every unit is an instance of exactly one item.
REPRESENTS_ISSUE	LIBRARY_UNIT $N : 1$ PERIODICAL_ISSUE	Optional on the unit side; used only for periodical issues.
BORROWED_BY	LOAN $N : 1$ MEMBER	Every loan has exactly one member.
USES	LOAN $N : 1$ LIBRARY_UNIT	Every loan uses exactly one unit.

Relationship	Multiplicity	Participation rule
PROCESSED_BY	LOAN $N : 1$ EMPLOYEE	At most one employee processes a loan; participation is optional.
CAUSED_BY	FINE $N : 1$ LOAN	Every fine is caused by exactly one loan.
SUPERVISES	EMPLOYEE $N : 1$ EMPLOYEE	An employee has at most one supervisor.
HELD_IN	EVENT $N : 1$ ROOM	Every event is held in exactly one room.
ORGANIZED_BY	EVENT $N : 1$ EMPLOYEE	Every event has exactly one organizer.
RECOMMENDED_FOR	EVENT $M : N$ AUDIENCE	Every event has at least one recommended audience.
FOR_EVENT	REGISTRATION $N : 1$ EVENT	Every registration has exactly one event.
FOR_MEMBER	REGISTRATION $N : 1$ MEMBER	Every registration has exactly one member.
SUBMITTED_BY	ITEM_PROPOSAL $N : 1$ MEMBER	Optional generally; required for a member donation or member request.
REVIEWED_BY	ITEM_PROPOSAL $N : 1$ EMPLOYEE	At most one reviewing employee.
RESULTS_IN	ITEM_PROPOSAL $1 : 1$ CATALOGUE_ITEM	Optional; used only after an accepted proposal has been catalogued.
APPLICANT	VOLUNTEER_APPLICATION $N : 1$ MEMBER	Every application has exactly one applicant.
REVIEWS	VOLUNTEER_APPLICATION $N : 1$ EMPLOYEE	At most one employee reviews an application.
SUPPORTS	VOLUNTEER_APPLICATION $N : 1$ EVENT	Optional; absent for general library work.
REQUESTED_BY	HELP_REQUEST $N : 1$ MEMBER	Every request has exactly one requesting member.
ASSIGNED_TO	HELP_REQUEST $N : 1$ EMPLOYEE	At most one employee is assigned.

9 Additional Semantic Constraints

Some business rules depend on dates, counts, sums, or status values and cannot be fully represented by arrows and participation lines in an E/R diagram. They remain part of the conceptual design and will later be enforced with SQL constraints, triggers, transactions, or application logic:

- Only an active member may borrow. A member may have at most five active loans and may not borrow while an item is overdue or unpaid fines total at least \$20.
- A physical copy may have at most one active loan. Concurrent digital loans for a digital licence may not exceed its licenceLimit.
- dueAt must be later than borrowedAt. A late physical return is charged \$0.25 per day, and all fine amounts are nonnegative.
- An event's endTime must be after its startTime. maximumAttendance may not exceed its room's capacity, and two events may not overlap in the same room.
- The pair (memberNo,eventNo) is unique in REGISTRATION. Registration is allowed only for a scheduled event with available capacity.
- A proposal can participate in RESULTS_IN only after it is accepted. Rejected proposals remain stored as history.

- A volunteer application can receive an assignment only after approval. A help request can become resolved only when response and completedAt are present.

10 Coverage of Required Application Actions

The model supports all eight required actions: catalogue and unit records allow item search; LOAN and FINE support borrowing and returning; ITEM_PROPOSAL supports donations and future acquisitions; EVENT, AUDIENCE, ROOM, and REGISTRATION support event search and registration; VOLUNTEER_APPLICATION supports volunteering; and HELP_REQUEST supports asking a librarian for help.

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Step 3: Anomaly and BCNF Analysis

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1 Purpose and Conclusion

This step translates the E/R model from Step 2 into relation schemas and examines the functional dependencies (FDs) that follow from the stated library semantics. A relation is in Boyce–Codd normal form (BCNF) exactly when, for every non-trivial FD $X \rightarrow Y$ that holds in the relation, X is a superkey.

Conclusion. Every relation in the final design is in BCNF. The E/R design already separates facts whose determinants are different, so no further BCNF decomposition is required. Therefore, the final schemas do not permit the insertion, deletion, or update anomalies caused by bad functional dependencies.

2 Relational Translation of the E/R Model

An underlined attribute is the selected primary key. “FK” denotes a foreign key, “AK” denotes an additional candidate key, and “optional” denotes an attribute that may be absent. Foreign-key references are stated after each group so that the schemas remain readable.

2.1 Members and Personnel

MEMBER(memberNo, fullName, email^{AK}, phone, mailingAddress, dateJoined,
status, expiryDate)

EMPLOYEE(employeeNo, fullName, email^{AK}, phone, jobTitle, hireDate,
employmentStatus, supervisorNo^{optional})

In EMPLOYEE, *supervisorNo* is a self-referencing foreign key to *employeeNo*. Its optional value implements the rule that an employee has at most one supervisor.

2.2 Catalogue, Issues, and Lendable Units

CREATOR(creatorID, creatorName)
MATERIALTYPE(typeName, defaultLoanDays)
CATALOGUEITEM(catalogueNo, title, publisher, publicationYear, language,
subject, description, standardID^{optional}, typeName^{FK})
CREATEDBY(catalogueNo, creatorID)
PERIODICALISSUE(catalogueNo, volume, issueNo, publicationDate)
LIBRARYUNIT(unitID, catalogueNo^{FK}, status)
PHYSICALCOPY(unitID, barcode^{AK}, shelfLocation, acquisitionDate, condition)
DIGITALLICENCE(unitID, licenceNo^{AK}, accessURL, licenceLimit)
REPRESENTSISSUE(unitID, catalogueNo^{FK}, volume^{FK}, issueNo^{FK})

CATALOGUEITEM.typeName references MATERIALTYPE. CREATEDBY references CATALOGUEITEM and CREATOR. PERIODICALISSUE references CATALOGUEITEM; its three underlined attributes form one composite key. LIBRARYUNIT references CATALOGUEITEM. The key of each subtype, PHYSICALCOPY and DIGITALLICENCE, is also a foreign key to LIBRARYUNIT. REPRESENTSISSUE references LIBRARYUNIT and the composite key of PERIODICALISSUE. Keeping this optional relationship in a separate table avoids null issue attributes for non-periodical units.

The *standardID* of a catalogue item is optional because some library materials have no ISBN, ISSN, or other standard identifier. When present, it must be unique. Because a classical candidate key must identify every tuple of a relation, an optional *standardID* is not treated as a candidate key or as the determinant of an ordinary FD over all of CATALOGUEITEM. The candidate key is *catalogueNo*. The conditional uniqueness of each non-null *standardID* is a separate integrity constraint for Step 4.

2.3 Loans and Fines

LOAN(loanID, memberNo^{FK}, unitID^{FK}, processedByEmployeeNo^{optional},
borrowedAt, dueAt, returnedAt^{optional})
FINE(fineID, loanID^{FK}, amount, assessedDate, reason, paymentStatus, paymentDate^{optional})

LOAN references MEMBER, LIBRARYUNIT, and optionally EMPLOYEE. FINE references LOAN. The loan foreign key is not unique in FINE, because the E/R model allows one loan to cause zero or more fines.

2.4 Events and Registrations

ROOM(roomNo, roomName, floor, capacity, accessibility, availabilityStatus)
EVENT(eventNo, title, eventType, description, startTime, endTime,
maximumAttendance, status, roomNo^{FK}, organizerEmployeeNo^{FK})
AUDIENCE(audienceName, description)
RECOMMENDEDFOR(eventNo, audienceName)
REGISTRATION(registrationID, eventNo^{FK}, memberNo^{FK}, registrationTime, status, attended)

EVENT references ROOM and EMPLOYEE. RECOMMENDEDFOR references EVENT and AUDIENCE. REGISTRATION references EVENT and MEMBER, and (*eventNo*, *memberNo*) is an additional candidate key. Thus, a member cannot register for the same event twice.

2.5 Proposals, Volunteering, and Help

ITEMPROPOSAL(proposalNo, title, creator, materialType, publicationInformation,
sourceType, dateSubmitted, condition^{optional}, notes, decisionStatus,
decisionDate^{optional}, memberNo^{optional}, reviewerEmployeeNo^{optional})

RESULTSIN(proposalNo, catalogueNo^{AK})

VOLUNTEERAPPLICATION(applicationNo, memberNo^{FK}, applicationDate, areaOfInterest,
availability, skills, status, assignmentNotes^{optional},
reviewerEmployeeNo^{optional}, eventNo^{optional})

HELPREQUEST(requestNo, memberNo^{FK}, category, questionDescription, submittedAt,
priority, status, response^{optional}, completedAt^{optional},
assignedEmployeeNo^{optional})

In ITEMPROPOSAL, *memberNo* and *reviewerEmployeeNo* reference MEMBER and EMPLOYEE. RESULTSIN references ITEMPROPOSAL and CATALOGUEITEM; both attributes are candidate keys because RESULTSIN is one-to-one. VOLUNTEERAPPLICATION references its applicant and optionally its reviewer and supported event. HELPREQUEST references its requesting member and optionally its assigned employee. A proposal's condition may be absent when it is not applicable; volunteer assignment notes may be absent before an assignment is made; and a help-request response and completion time may be absent until the request is resolved.

3 Functional-Dependency Analysis

3.1 Method and Assumptions

For each relation below, a minimal semantic FD basis is given. All other non-trivial FDs follow from this basis by Armstrong's axioms. Writing the minimal basis avoids listing redundant augmented forms such as $\{memberNo, phone\} \rightarrow fullName$ when $memberNo \rightarrow fullName$ already holds.

The following assumptions are part of the data semantics:

- member and employee email addresses are required and unique within their respective relations;
- barcodes and licence numbers are required and unique within their respective subtype relations;
- a standard identifier is unique whenever one is present; because it is optional, this is treated as a conditional uniqueness constraint rather than an ordinary FD over all catalogue-item tuples;
- names, titles, phone numbers, dates, status values, room names, and descriptive fields are not assumed to be unique;
- historical records are retained, so a member, unit, employee, room, or event may occur in many related tuples; and
- no FD is inferred merely from an accidental pattern in the current sample data. An FD is included only if it must hold in every legal database state under the stated business rules.

In the table, $K \rightarrow \text{rest}$ means that K determines every other attribute of that relation. This is a compact representation of the individual non-trivial dependencies having the same determinant.

3.2 Compact FD Basis, Candidate Keys, and BCNF Test

Relation	Minimal non-trivial FD basis	Candidate key(s)	BCNF test
MEMBER	$memberNo \rightarrow rest; email \rightarrow rest$	$\{memberNo\}, \{email\}$	Both determinants are keys.
EMPLOYEE	$employeeNo \rightarrow rest; email \rightarrow rest$	$\{employeeNo\}, \{email\}$	Both determinants are keys.
CREATOR	$creatorID \rightarrow creatorName$	$\{creatorID\}$	The determinant is a key.
MATERIALTYPE	$typeName \rightarrow defaultLoanDays$	$\{typeName\}$	The determinant is a key.
CATALOGUEITEM	$catalogueNo \rightarrow rest$	$\{catalogueNo\}$	The determinant is a key.
CREATEDBY	No non-trivial FDs. Both columns are required to identify one association.	$\{catalogueNo, creatorID\}$	BCNF holds vacuously.
PERIODICALISSUE	$\{catalogueNo, volume, issueNo\} \rightarrow publicationDate$	$\{catalogueNo, volume, issueNo\}$	The determinant is the key.
LIBRARYUNIT	$unitID \rightarrow \{catalogueNo, status\}$	$\{unitID\}$	The determinant is a key.
PHYSICALCOPY	$unitID \rightarrow rest; barcode \rightarrow rest$	$\{unitID\}, \{barcode\}$	Both determinants are keys.
DIGITALLICENCE	$unitID \rightarrow rest; licenceNo \rightarrow rest$	$\{unitID\}, \{licenceNo\}$	Both determinants are keys.
REPRESENTSISSUE	$unitID \rightarrow \{catalogueNo, volume, issueNo\}$	$\{unitID\}$	The determinant is a key.
LOAN	$loanID \rightarrow rest$	$\{loanID\}$	The determinant is a key.
FINE	$fineID \rightarrow rest$	$\{fineID\}$	The determinant is a key.
ROOM	$roomNo \rightarrow rest$	$\{roomNo\}$	The determinant is a key.
EVENT	$eventNo \rightarrow rest$	$\{eventNo\}$	The determinant is a key.
AUDIENCE	$audienceName \rightarrow description$	$\{audienceName\}$	The determinant is a key.
RECOMMENDEDFOR	No non-trivial FDs. Both columns are required to identify one association.	$\{eventNo, audienceName\}$	BCNF holds vacuously.
REGISTRATION	$registrationID \rightarrow rest; \{eventNo, memberNo\} \rightarrow rest$	$\{registrationID\}, \{eventNo, memberNo\}$	Both determinants are keys.
ITEMPROPOSAL	$proposalNo \rightarrow rest$	$\{proposalNo\}$	The determinant is a key.
RESULTSIN	$proposalNo \rightarrow catalogueNo; catalogueNo \rightarrow proposalNo$	$\{proposalNo\}, \{catalogueNo\}$	Both determinants are keys.
VOLUNTEERAPPLICATION	$applicationNo \rightarrow rest$	$\{applicationNo\}$	The determinant is a key.
HELPREQUEST	$requestNo \rightarrow rest$	$\{requestNo\}$	The determinant is a key.

4 Why the Foreign Keys Do Not Create Additional Bad FDs

The arrows in the E/R diagram create several many-to-one relationships. The foreign key on the “many” side is determined by that relation’s key, but the reverse dependency generally does not hold. In particular:

- $memberNo \not\rightarrow loanID$ because one member can have many loans;
- $unitID \not\rightarrow loanID$ because one unit can be borrowed repeatedly over time;
- $loanID \not\rightarrow fineID$ because a loan may cause more than one fine;
- $roomNo \not\rightarrow eventNo$ and $organizerEmployeeNo \not\rightarrow eventNo$ because a room and an employee may be associated with many events;
- $memberNo \not\rightarrow applicationNo$ and $memberNo \not\rightarrow requestNo$ because a member may submit many volunteer applications and help requests; and
- $typeName \rightarrow defaultLoanDays$ is stored only in MATERIALTYPE, not in CATALOGUEITEM. Likewise, $roomNo \rightarrow capacity$ is stored only in ROOM, and member details are stored only in MEMBER.

Consequently, foreign keys do not duplicate descriptive facts in referencing relations. They are inclusion constraints, while the descriptive FDs remain inside the relations whose keys determine those facts.

5 Formal BCNF Proof

Let R be any relation in the final design and let $X \rightarrow Y$ be a non-trivial FD that follows from the library semantics.

1. If R is a regular entity or relationship relation with descriptive attributes, the FD-basis table shows that every minimal determinant is its primary key or an alternate candidate key. Hence X contains a candidate key and is a superkey.
2. If R is CREATEDBY or RECOMMENDEDFOR, there is no non-trivial FD. Each tuple consists only of the two attributes forming the many-to-many association key, so BCNF holds vacuously.
3. If R is RESULTSIN, either attribute determines the other due to the one-to-one constraint. Each determinant is therefore a candidate key.
4. The composite determinant of PERIODICALISSUE is its full key, and the determinant $unitID$ in REPRESENTSISSUE is its key.

Thus, in every possible case, the left side of each non-trivial FD is a superkey. By the definition of BCNF, every relation in the final schema is in BCNF.

6 Anomaly Check

The absence of bad FDs prevents the three anomaly types discussed in class:

- **Update anomalies:** changing a material type’s default loan period, a room’s capacity, a member’s email, or a creator’s name requires one update in its own relation rather than many updates in loan, event, or item tuples.
- **Insertion anomalies:** a new material type, creator, room, audience, member, or future-item proposal can be recorded without first creating a loan, event, registration, or catalogue copy.
- **Deletion anomalies:** deleting a returned loan, cancelled registration, event recommendation, or withdrawn physical copy does not erase the independent member, event, audience, creator, or catalogue facts.

For comparison, a single wide relation containing loan, member, unit, catalogue, and material-type data would contain the bad dependencies

$$\begin{array}{ll} memberNo \rightarrow \text{member details}, & unitID \rightarrow catalogueNo, \\ catalogueNo \rightarrow \{\text{item details}, typeName\}, & typeName \rightarrow defaultLoanDays. \end{array}$$

Those determinants would not be superkeys of the wide relation. The final design avoids this problem by storing the facts in MEMBER, LIBRARYUNIT, CATALOGUEITEM, and MATERIALTYPE, respectively, with LOAN retaining only their foreign keys.

7 Constraints Beyond Functional Dependencies

BCNF prevents anomalies caused by functional dependencies, but it does not by itself enforce every business rule. The following are not bad FDs and will be implemented in Step 4 using keys, foreign keys, CHECK constraints, triggers, transactions, or application logic:

- at most five active loans per member, no borrowing by an inactive member, and borrowing restrictions for overdue items or unpaid fines;
- at most one active loan for a physical copy and at most *licenceLimit* simultaneous loans for a digital licence;
- chronological conditions such as *dueAt* > *borrowedAt* and *endTime* > *startTime*;
- no overlapping events in the same room and attendance not exceeding the event or room capacity;
- total participation requirements such as every catalogue item having at least one creator and every event having at least one audience; and
- conditional uniqueness of each non-null *standardID* and the requirement that the catalogue number in a REPRESENTS ISSUE tuple agree with the catalogue number of its referenced LIBRARYUNIT; and
- conditional rules for accepted proposals, approved volunteer assignments, resolved help requests, and subclass disjointness and completeness.

These integrity constraints complement the BCNF design; they do not require further decomposition.